

The visual representation of different types of measures for self-similar sets

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Self-similar set & measure

Let $m > 0$ be an integer and $\{f_j(x) = \rho_j x + d_j, |\rho_j| < 1, d_j \in \mathbb{R}^n\}_{j=1}^m$ be an *iterated function system (IFS)* on $\mathbb{R}^n$.

- There exists a unique non-empty compact set K (the *attractor* or *self-similar set*) that satisfies

$$K = \bigcup_{j=1}^m f_j(K).$$

- For any probability vector $\mathbf{P} = (p_1, p_2, \dots, p_m)$, there is a unique probability measure $\sigma = \sigma_{\mathbf{P}}$ (the *self-similar measure*) on K such that

$$\sigma = \sum_{j=1}^m p_j \sigma \circ f_j^{-1}.$$

Let $\mathcal{M}$ be a family of subsets in $\mathbb{R}^n$. For $E \subseteq \mathbb{R}^n$, $s \geq 0$ and $\delta > 0$, let

$$\mathcal{W}_{\mathcal{M},\delta}^s(E) = \inf \left\{ \sum_{i=1}^{\infty} (\text{diam } U_i)^s : E \subseteq \bigcup_{i=1}^{\infty} U_i, \right. \\ \left. U_i \in \mathcal{M}, \text{diam } U_i \leq \delta \text{ for all } i \right\}.$$

The s -dimensional $\mathcal{M}$ -Hausdorff type measure of a subset $E \subseteq \mathbb{R}^n$ is defined by

$$\mathcal{W}_{\mathcal{M}}^s(E) = \lim_{\delta \rightarrow 0} \mathcal{W}_{\mathcal{M},\delta}^s(E).$$

- Take

$$\mathcal{M} = \{U \subseteq \mathbb{R}^n \mid U \text{ is an open (or a closed) set}\},$$

then $\mathcal{W}_{\mathcal{M}}^s(E)$ equals s -dimensional Hausdorff measure $\mathcal{H}^s$.

- Particularly, take

$$\mathcal{M} = \mathcal{M}_B = \{B \subseteq \mathbb{R}^n \mid B \text{ is a ball}\},$$

then $\mathcal{W}_{\mathcal{M}}^s(E)$ equals s -dimensional spherical measure $\mathcal{S}^s$.

Let $\delta > 0$, $\mathcal{B} := \{B(x_i, r_i)\}_{i=1}^{\infty}$ be a family of balls of $\mathbb{R}^n$.

- $\mathcal{B}$ is a **centred δ -covering** of E if $r_i \leq \delta$ and $x_i \in E$ for each i and $E \subset \bigcup_{i=1}^{\infty} B(x_i, r_i)$.
- $\mathcal{B}$ is a **packing** of E if $x_i \in E$, $r_i \leq \delta$ and these balls are disjoint.

For $E \subset \mathbb{R}^n$, $s \geq 0$ and $\delta > 0$, let

$$\mathcal{C}_{\delta}^s(E) = \inf \left\{ \sum_{i=1}^{\infty} (\text{diam } B_i)^s : \{B_i\}_{i=1}^{\infty} \text{ is a centered } \delta\text{-covering of } E \right\},$$

$$\mathcal{P}_{\delta}^s(E) = \sup \left\{ \sum_{i=1}^{\infty} (\text{diam } B_i)^s : \{B_i\}_{i=1}^{\infty} \text{ is a } \delta\text{-packing of } E \right\}.$$

Since $\mathcal{C}_\delta^s(E)$ and $\mathcal{P}_\delta^s(E)$ decrease with δ , the limits

$$\mathcal{C}_0^s(E) = \lim_{\delta \rightarrow 0} \mathcal{C}_\delta^s(E) \quad \text{and} \quad \mathcal{P}_0^s(E) = \lim_{\delta \rightarrow 0} \mathcal{P}_\delta^s(E)$$

exist (possibly 0 or ∞).

Note that $\mathcal{C}_0^s$ is **not monotone** and $\mathcal{P}_0^s$ is **not countable subadditive**. But they both give rise to measures, called respectively the *packing measure* and *Hausdorff centered measure*, as follows:

$$\mathcal{P}^s(E) = \inf \left\{ \sum_{i=1}^{\infty} \mathcal{P}_0^s(E_i) : E \subset \bigcup_{i=1}^{\infty} E_i \right\},$$
$$\mathcal{C}^s(E) = \sup \{ \mathcal{C}_0^s(F), F \subset E \}.$$

For $E \subseteq \mathbb{R}^n$, $s \geq 0$, we have

- $\mathcal{H}^s(E) \leq \mathcal{S}^s(E) \leq \mathcal{C}^s(E) \leq \mathcal{P}^s(E)$.
- $\mathcal{H}^s(E) \leq \mathcal{S}^s(E) \leq 2^s \mathcal{H}^s(E)$.
- $\mathcal{H}^s(E) \leq \mathcal{C}^s(E) \leq 2^s \mathcal{H}^s(E)$.

Hasudorff measures

- Marion 1985: linear Cantor sets satisfying the OSC.
- Ayer-Strichartz 1999: the self-similar Cantor sets satisfying weaker conditions than the OSC.
- ...
- Some estimates on the Sierpinski carpet and Sierpinski gasket (Jia-Zhou-Zhu 2002; Jia 2009, etc.)

Packing measures

- Olsen 2008: based on density theorem, satisfying the SSC.
- Develop by Moran 2009; Qiu 2012, etc.
- Feng 2003, Feng-Zhou 2004; Zuberger 2019, etc.
- Some estimates on higher dimension.

Separation condition

We recall some separation conditions.

- The IFS satisfies the **open set condition (OSC)** if there exists a non-empty bounded open set V such that

$$\bigcup_{j=1}^m f_j(V) \subset V \text{ and } f_i(V) \cap f_j(V) = \emptyset \text{ for } i \neq j.$$

- The IFS satisfies **the strong open set condition (SOSC)**, if, in addition, $V \cap K \neq \emptyset$.
- The IFS satisfies **the strong separation condition (SSC)** if

$$f_i(K) \cap f_j(K) = \emptyset \text{ for } i \neq j.$$

The OSC is the most important separation condition.

- $\text{SSC} \Rightarrow \text{SOSC}$.
- $\text{SOSC} \Leftrightarrow \text{OSC}$ for the self-similar IFSs.
- (Schief 94) The self-similar IFSs

$$\text{OSC} \Leftrightarrow 0 < \mathcal{H}^s(K) < \infty.$$

Corollary (Fu-Z.-Qiu)

Let $\{f_j\}_{j=1}^m$ be a self-similar IFS and let K be the corresponding self-similar set. Then

$$\text{OSC} \Leftrightarrow 0 < \mathcal{P}^s(K) < \infty.$$

Compute the measures of self-similar set $K(A, \mathcal{D})$

In the following, we consider the pair $(A, \mathcal{D})$, where

- $A \in M_n(\mathbb{R})$ is an expanding similarity matrix with scaling factor $\rho > 1$.
- $\mathcal{D} \subset \mathbb{R}^n$ and $\#\mathcal{D} < |\det(A)|$.

self-similar set K : $AK = K + \mathcal{D}$.

Our goal is to compute the different measures $(\mathcal{H}^s, \mathcal{S}^s, \mathcal{C}^s, \mathcal{P}^s)$ of K .
An appropriate notion of density for the discrete measure is defined by

$$\mu = \lim_{N \rightarrow \infty} \sum_{d_0, d_1, \dots, d_{N-1} \in \mathcal{D}} \delta_{d_0 + Ad_1 + \dots + A^{N-1}d_{N-1}}. \quad (1)$$

Upper convex and upper spherical density of ν

For a Borel measure ν on $\mathbb{R}^n$, the *s-dimensional upper convex density* and *s-dimensional upper spherical density* of ν are defined by

$$\Theta_{s,conv}^+(\nu) = \limsup_{r \rightarrow \infty} \sup_{\substack{\text{diam}(U) \geq r > 0 \\ U \text{ is convex}}} \frac{\nu(U)}{(\text{diam } U)^s}$$

and

$$\Theta_{s,sph}^+(\nu) = \limsup_{r \rightarrow \infty} \sup_{\substack{\text{diam}(B) \geq r > 0 \\ B \text{ is a ball}}} \frac{\nu(B)}{(\text{diam } B)^s}$$

respectively.

For a Borel measure ν on $\mathbb{R}^n$, the *s -dimensional upper Beurling density* and *s -dimensional lower Beurling density* of ν are defined respectively by

$$\Theta_s^+(\nu) = \limsup_{r \rightarrow \infty} \sup_{x \in \mathbb{R}^n} \frac{\nu(B(x, r))}{(2r)^s}$$

and

$$\Theta_s^-(\nu) = \liminf_{r \rightarrow \infty} \sup_{x \in \mathbb{R}^n} \frac{\nu(B(x, r))}{(2r)^s}.$$

If they agree, their common value is called the *s -dimensional Beurling density* of ν , and denoted by $\Theta_s(\nu)$.

Theorem (Fu-Z.-Qiu)

Suppose that K is the self-similar set determined by the pair $(A, \mathcal{D})$, and $s = \log_\rho(\#D)$. Then

$$\mathcal{H}^s(K) = 1/\Theta_{s,\text{conv.}}^+(\mu),$$

$$\mathcal{S}^s(K) = 1/\Theta_{s,\text{sph.}}^+(\mu),$$

$$\mathcal{C}^s(K) = 1/\Theta_s^+(\mu),$$

$$\mathcal{P}^s(K) = 1/\Theta_s^-(\mu),$$

where μ is given by (1), $\Theta_{s,\text{conv.}}^+(\mu)$, $\Theta_{s,\text{sph.}}^+(\mu)$, $\Theta_s^+(\mu)$ and $\Theta_s^-(\mu)$ are given respectively by

$$\Theta_{s, \text{conv.}}^+(\mu) = \limsup_{r \rightarrow \infty} \sup_{\substack{\text{diam}(U) \geq r > 0 \\ U \text{ is convex}}} \frac{\mu(U)}{(\text{diam } U)^s}, \quad \hookrightarrow (\mathcal{H}^s)$$

$$\Theta_{s, \text{sph.}}^+(\mu) = \limsup_{r \rightarrow \infty} \sup_{\substack{\text{diam}(B) \geq r > 0 \\ B \text{ is a ball}}} \frac{\mu(B)}{(\text{diam } B)^s}, \quad \hookrightarrow (\mathcal{S}^s)$$

$$\Theta_s^+(\mu) = \limsup_{r \rightarrow \infty} \sup_{x \in \mathbb{R}^n} \frac{\mu(B(x, r))}{(2r)^s}, \quad \hookrightarrow (\mathcal{C}^s)$$

$$\Theta_s^-(\mu) = \liminf_{r \rightarrow \infty} \sup_{x \in \mathbb{R}^n} \frac{\mu(B(x, r))}{(2r)^s}. \quad \hookrightarrow (\mathcal{P}^s)$$

We point out that the conclusion “ $\mathcal{H}^s(K) = 1/\Theta_{s,c}^+(\mu)$ ” was first proved by Fu and Gabardo (Constr. Approx, 2014).

Lemma (Olsen, Mattila, etc)

Let $E \subseteq \mathbb{R}^n$, $s > 0$ and let E be an s -set w.r.t. $\mathcal{H}^s, \mathcal{S}^s, \mathcal{C}^s, \mathcal{P}^s$. Then

$$D_{s, \text{conv.}}^*(\mathcal{H}^s \upharpoonright E, x) = \limsup_{r \rightarrow 0} \sup_{\substack{0 < \text{diam}(U) \leq r, x \in U \\ U \text{ is a convex}}} \frac{(\mathcal{H}^s \upharpoonright E)(U)}{(\text{diam } U)^s} = 1,$$

$$D_{s, \text{sph.}}^*(\mathcal{S}^s \upharpoonright E, x) = \limsup_{r \rightarrow 0} \sup_{\substack{0 < \text{diam}(B) \leq r, x \in B \\ B \text{ is a ball}}} \frac{(\mathcal{S}^s \upharpoonright E)(B)}{(\text{diam } B)^s} = 1,$$

$$D_s^*(\mathcal{C}^s \upharpoonright E, x) = \limsup_{r \rightarrow 0} \frac{(\mathcal{C}^s \upharpoonright E)(B(x, r))}{(2r)^s} = 1,$$

$$D_{s,*}(\mathcal{P}^s \upharpoonright E, x) = \liminf_{r \rightarrow 0} \frac{(\mathcal{P}^s \upharpoonright E)(B(x, r))}{(2r)^s} = 1,$$

for $\mathcal{H}^s, \mathcal{S}^s, \mathcal{C}^s, \mathcal{P}^s$ -almost all $x \in E$ respectively.

Lemma (Fu-Z.-Qiu)

Let ν and σ be two positive Borel measures on $\mathbb{R}^n$ with σ being a probability measure. Then

- (i) $\Theta_{s,\text{conv.}}^+(\nu * \sigma) = \Theta_{s,\text{conv.}}^+(\nu)$ and $\Theta_{s,\text{sph.}}^+(\nu * \sigma) = \Theta_{s,\text{sph.}}^+(\nu)$.
- (ii) $\Theta_s^+(\nu * \sigma) = \Theta_s^+(\nu)$ and $\Theta_s^-(\nu * \sigma) = \Theta_s^-(\nu)$.
- (iii) For the pair $(A, \mathcal{D})$ and any Borel measurable set $E \subseteq \mathbb{R}^n$, if we take $\nu = \mu_N = \sum_{d_0, d_1, \dots, d_{N-1} \in \mathcal{D}} \delta_{d_0 + Ad_1 + \dots + A^{N-1}d_{N-1}}$, then we have

$$\sigma(A^{-N}E) = \frac{1}{m^N}(\mu_N * \sigma)(E), \text{ where } m = \#\mathcal{D}.$$

Proof of $\mathcal{P}^s(K) = (\Theta_s^-(\mu))^{-1}$ for not OSC

WLOG, suppose that $0 \in \mathcal{D}$. Let

$$\mathcal{D}_M = \left\{ \sum_{j=0}^{M-1} A^j d_j : d_j \in \mathcal{D} \right\} \quad \text{and} \quad \mathcal{D}_\infty = \bigcup_{M \geq 1} \mathcal{D}_M.$$

The IFS $\{f_d\}_{d \in \mathcal{D}}$ satisfies the OSC $\Leftrightarrow \#\mathcal{D}_M = (\#\mathcal{D})^M$ and $\mathcal{D}_\infty$ is a uniformly discrete set, i.e., there exists $\eta > 0$ such that $\|x - y\| \geq \eta$ for any $x \neq y \in \mathcal{D}_\infty$ (He-Lau, 08).

- Not OSC $\Rightarrow \mathcal{P}^s(K) = 0 \Rightarrow (\#\mathcal{D})^M$ expansions in $\mathcal{D}_M$ are not distinct or $\mathcal{D}_\infty$ is not a uniformly discrete set $\Rightarrow \Theta_s^-(\mu) = \infty$.

Summary

Local densities are the main tool used by Olsen etc.

$$\mathcal{H}^s(K) = \inf_{\substack{U \text{ is convex} \\ K \cap U \neq \emptyset}} \frac{(\text{diam } U)^s}{\sigma(U)} \quad \text{and} \quad \mathcal{P}^s(K) = \sup_{\substack{x \in K \\ 0 < r \leq r_0}} \frac{(2r)^s}{\sigma(B(x, r))}.$$

Our results:

$$\liminf_{r \rightarrow \infty} \sup_{x \in \mathbb{R}^n} \frac{\mu(B(x, r))}{(2r)^s} \leq \limsup_{r \rightarrow \infty} \sup_{x \in \mathbb{R}^n} \frac{\mu(B(x, r))}{(2r)^s} \leq \limsup_{r \rightarrow \infty} \sup_{\substack{\text{diam}(B) \geq r > 0 \\ B \text{ is a ball}}} \frac{\mu(B)}{(\text{diam } B)^s} \leq \limsup_{r \rightarrow \infty} \sup_{\substack{\text{diam}(U) \geq r > 0 \\ U \text{ is a convex}}} \frac{\mu(U)}{(\text{diam } U)^s}.$$

It is clear to see that $\mathcal{H}^s(K) \leq \mathcal{S}^s(K) \leq \mathcal{C}^s(K) \leq \mathcal{P}^s(K)$.

- It can also be represented when IFS does **not satisfy** the OSC.
- Our method can also be extended to incorporate **pseudo-norm on self-affine sets**, enabling us to estimate the **pseudo-packing (Hausdorff centered) measures** (see Fu-Gabardo-Qiu, 2020).

Computing the exact $\mathcal{H}^s$ and $\mathcal{P}^s$ is a very difficult task

Example: Let $\mathcal{C}_{1/3}$ be the classical one-third Cantor set. Then $\mathcal{H}^{\log_3^2}(\mathcal{C}_{1/3}) = 1$, $\mathcal{P}^{\log_3^2}(\mathcal{C}_{1/3}) = 4^{\log_3^2} \approx 2.39$.

We are interested in which self-similar sets K satisfying $0 < \mathcal{H}^s(K) = \mathcal{P}^s(K) < \infty$.

A set $E \subseteq \mathbb{R}^n$ is called **s -rectifiable** if there exists Lipschitz maps $\psi_i : \mathbb{R}^s \rightarrow \mathbb{R}^n$, $i = 1, 2, \dots$, and bounded subsets A_i of $\mathbb{R}^s$ such that

$$\mathcal{H}^s \left(E \setminus \bigcup_{i=1}^{\infty} \psi_i(A_i) \right) = 0.$$

A set $E \subseteq \mathbb{R}^n$ is **purely s -unrectifiable** if $\mathcal{H}^s(E \cap F) = 0$ for every s -rectifiable set $F \subset \mathbb{R}^n$.

Theorem (Mattila)

For a set E to be an s -rectifiable set if and only if

$$\Phi^s(E, x) = \lim_{r \rightarrow 0} \frac{\mathcal{H}^s(E \cap B(x, r))}{(2r)^s}$$

exists and equals 1 for $\mathcal{H}^s$ -almost all $x \in E$.

An s -set $E \subset \mathbb{R}^n$ can be decomposed as $E = E_R \cup E_I$, where E_R is **regular**, that is, with

$$\limsup_{r \rightarrow 0} \frac{\mathcal{H}^s(E_R \cap B(x, r))}{(2r)^s} = \liminf_{r \rightarrow 0} \frac{\mathcal{H}^s(E_R \cap B(x, r))}{(2r)^s} = 1,$$

for $\mathcal{H}^s$ -almost all $x \in E_R$, and E_I is **irregular**, with

$$\liminf_{r \rightarrow 0} \frac{\mathcal{H}^s(E_I \cap B(x, r))}{(2r)^s} \leq c \limsup_{r \rightarrow 0} \frac{\mathcal{H}^s(E_I \cap B(x, r))}{(2r)^s}$$

for $\mathcal{H}^s$ -almost all $x \in E_I$, where $c < 1$ depends only on n and s .

- An s -rectifiable set $\Leftrightarrow$ a regular set.

Lemma (Raymond and Tricot)

Let $0 < s < n$ and $E \subset \mathbb{R}^n$ with $0 < \mathcal{P}^s(E) < \infty$. Then
 $\mathcal{P}^s(E) = \mathcal{H}^s(E) \Leftrightarrow s$ is an integer and $\mathcal{P}^s \upharpoonright E$ is s -rectifiable.

Using the fact: if $\mathcal{P}^s \upharpoonright E$ is s -rectifiable, then $\mathcal{P}^s \upharpoonright E \ll \mathcal{H}^s$.

$\Rightarrow$ A set $E \subset \mathbb{R}^n$ with $0 < \mathcal{P}^s(E) = \mathcal{H}^s(E) < \infty$ if and only if E is a regular set and $s \in \mathbb{N}$.

Besicovitch projection theorem

- proj_θ denotes orthogonal projection onto the line L_θ that makes angle θ with some axis

Theorem (Besicovitch projection theorem)

Let E be a subset in $\mathbb{R}^2$.

- (i) If E is a regular 1-set, then $\mathcal{L}^1(\text{proj}_\theta(E)) > 0$ for all but at most one value of $\theta \in [0, \pi)$.
- (ii) If E is an irregular 1-set, then $\mathcal{L}^1(\text{proj}_\theta(E)) = 0$ for almost all $\theta \in [0, \pi)$.

Corollary

A 1-set E is irregular if and only if it has projections of Lebesgue measure zero in two distinct directions.

Bedford-McMullen carpet

Let $n, m \geq 2$ and let $S \subset \{0, 1, \dots, n-1\} \times \{0, 1, \dots, m-1\}$. Divide the unit square $[0, 1]^2$ into subrectangles of size $\frac{1}{n} \times \frac{1}{m}$ and keep only the squares

$$\left[\frac{i}{n}, \frac{i+1}{n} \right] \times \left[\frac{j}{m}, \frac{j+1}{m} \right],$$

where $(i, j) \in S$. We iterate the procedure. The resulting set is the carpet. It can also be written as

$$K = \left\{ \left(\sum_{k=1}^{\infty} \frac{x_k}{n^k}, \sum_{k=1}^{\infty} \frac{y_k}{m^k} \right) : (x_k, y_k) \in S, \forall k \geq 1 \right\}.$$

Example

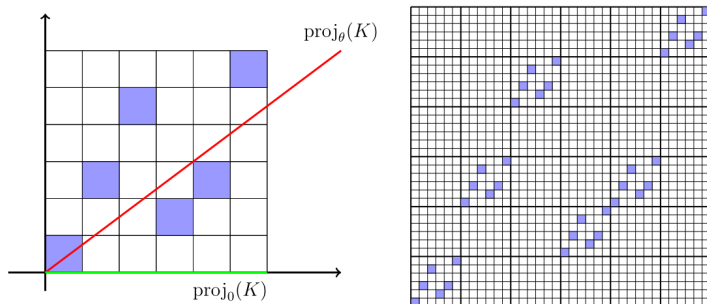


FIGURE 5.1. Illustrate of projection on some line on first iterated on the left hand side. And the right hand is the second iterated.

Theorem (Federer 1947, White 1998)

Let $E \subset \mathbb{R}^n$ be a $\mathcal{H}^s$ -measurable set with $0 < \mathcal{H}^s(E) < \infty$. Then E is an irregular set if and only if $\mathcal{L}^s(\text{proj}_\Sigma(E)) = 0$ for almost all $\Sigma \in G(n, s)$, where $G(n, s)$ denotes all s -dimensional linear subspaces of $\mathbb{R}^n$.

Example in $\mathbb{R}^3$

Let $A = \text{diag}[2, 2, 2]$ and $D = \left\{ \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 10 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 16 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} \right\}$.

Then $0 < \mathcal{P}^2(K(A, D)) = \mathcal{H}^2(K(A, D)) < \infty$.

A MATLAB visual of projection

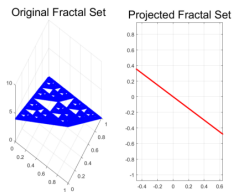


FIGURE 5.4. Normal vector $(3,4,0)$ of the cutting plane

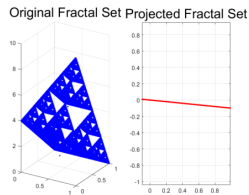


FIGURE 5.5. Normal vector $(2,20,0)$ of the cutting plane

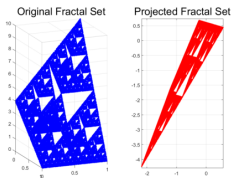


FIGURE 5.6. Normal vector $(3,4,5)$ of the cutting plane

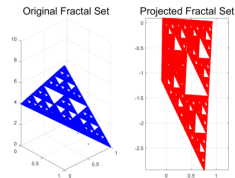


FIGURE 5.7. Normal vector $(2,25,8)$ of the cutting plane

Theorem (Fu-Z.-Qiu)

Let $\rho > 1$ be a positive integer number, $A = \text{diag}[\rho, \dots, \rho]$. If we choose digit set D satisfies $\mathbb{N} \ni s = \log_{\rho} \#D$ and such that the corresponding self-similar set K contains some s -dimensional coordinate-cube. Then

$$0 < \mathcal{P}^s(K) = \mathcal{H}^s(K) < \infty.$$

In particular, the s -dimensional coordinate-cube can be taken as the complete residual system of the s components corresponding to the matrix entries.

Further questions

Given a digit set $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ and expansive matrices A_j ($1 \leq j \leq N$), define

$$f_j(x) = A_j^{-1}(x + d_j).$$

How to compute the Hausdorff measures and packing measures of the corresponding self-similar sets?

Thank you!